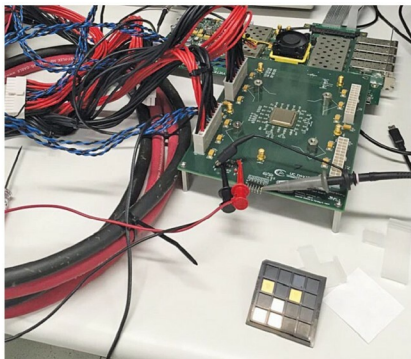


TECH NEWS

TECHNOLOGY UPDATES

World's first 1000-core processor



The microchip with 1000-processor cores is thought to be the fastest designed in a university lab (Image courtesy: www.ucdavis.edu)

Scientists have unveiled the world's first 1000-processor microchip, called KiloCore, which is capable of working through 1.78 trillion instructions every second. Each of those 1000 processors can run independently, which makes the chip suited to intensive tasks like encryption and weather forecasting. With some low-energy engineering, it can be powered by only one AA battery.

Because each processor can turn on and shut down independently of the others, it is about 100 times more efficient at executing instructions than today's laptops, according to the researchers who developed it at University of California, Davis, USA, and IBM.

Today's top-end smartphones have quad- or octa-core processors in them, so this is more than a 100-fold increase in terms of the number of cores. The more the cores, the more tasks you can get through simultaneously. While that does not make much of a difference for things like Facebook or Spotify, it can significantly boost supercomputer-level processing.

World's fastest supercomputer is made in China

Sunway TaihuLight, the world's fastest supercomputer, is entirely made in China and does not rely on American processors. Instead, it relies on the locally-made Shen-Wei processor and custom interconnect called Sunway Network. The supercomputer establishes Chinese dominance on supercomputing technology.

Sunway TaihuLight is currently functional at National Supercomputing Center at Wuxi near Shanghai, China. Its primary job is to help with top-notch engineering and research in climate and weather-related calculations, advanced biological and geological research and in manufacturing and Big Data analytics.

SW26010 is a 260-core microchip that cranks a little over three teraflops. TaihuLight has one such processor in each of the 40,960 nodes, allowing the machine to reach 125-peak petaflops. Linpack mark leaves out a couple of petaflops to mark 93 petaflops performance.



At its peak, the supercomputer can perform around 93,000 trillion calculations per second (Image courtesy: www.bbc.com)